



Tianhua Gao

Ph.D. in Engineering

Personal Website: <https://tianhuagao.github.io/>

| About Me

Tianhua Gao received the Ph.D. degree from the Graduate School of Systems and Information Engineering, University of Tsukuba, Japan, in March 2026. He was awarded the University of Tsukuba Fellowship and the JST SPRING Scholarship. He was also a Research Assistant at the Intelligent Systems Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Japan. His research interests focus on integrating control theory and neuroscience to advance the field of machine intelligence and control.

| Academic & Professional Memberships

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, IEEE Control Systems Society (CSS)
- Member, IEEE Robotics and Automation Society (RAS)
- Member, IEEE Industrial Electronics Society (IES)
- Member, American Society of Mechanical Engineers (ASME)
- Member, American Mathematical Society (AMS)

| Awards

- SPRING Scholarship 2023-2026
¥8,100,000 grant for PhD research. Support for Pioneering Research Initiated by the Next Generation (SPRING).
- University of Tsukuba Fellowship 2023
Full tuition coverage provided during the Ph.D. program.

| Publications

- J. Yu, J. Li and **T. Gao*** "Trajectory Planning for a Multi-UAV Rigi-Payload Cascaded Transportation System Based on Enhanced Tube-RRT*," in *IEEE Transactions on Aerospace and Electronic Systems (TAES)*, 2026 (Under Review).
- **T. Gao**, M. Izumita, K. Tomita and A. Kamimura, "A Sliced Learning Framework for Online Disturbance Identification in Quadrotor SO(3) Attitude Control," in *IEEE/ASME Transactions on Mechatronics (TMECH)*, 2026.
- **T. Gao**, K. Tomita and A. Kamimura, "Online Identification using Adaptive Laws and Neural Networks for MultiQuadrotor Centralized Transportation System," *Engineered Science*, 2025.
- **T. Gao**, K. Tomita and A. Kamimura, "Robustness Enhancement for Multi-Quadrotor Centralized Transportation System via Online Tuning and Learning," *American Control Conference (ACC)*, Denver, CO, USA, 2025.
- **T. Gao**, X. Han, K. Tomita and A. Kamimura, "Numerical Simulation of a Novel Cargo Handling Strategy: Using a Centralized Cable-Linked Dual-Multirotor System," *IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, Boston, MA, USA, 2024, pp. 345-352 .
- **T. Gao**, X. Han, K. Tomita and A. Kamimura, "Nonlinear Model-Assisted Control for Autonomous Parachute Precision Landing Recovery System," *61st Annual Conference of the Society of Instrument and Control*

| Technical Background

Mathematics:

- Topology
- Differential geometry and differential manifolds
- Lie groups and Lie algebras
- Graph theory and matrix theory
- Fiber bundle theory

Physics:

- Newtonian mechanics
- Lagrangian mechanics
- Hamiltonian mechanics
- Analytical mechanics

Control Theory:

- Classical control (PID)
- Optimal control (LQR, MPC, NMPC)
- Robust and adaptive control (ADRC, MRAC, L1 adaptive control)
- Nonlinear control and geometric control
- SO(3)-based attitude control and SE(3) coupled control
- Learning-based and neural network control
- Convex and non-convex optimization
- System identification and online learning

Uncrewed Aerial Vehicles (UAVs):

- ArduPilot (APM) and PX4 flight control systems
- Software-in-the-loop (SITL) simulation
- Gazebo simulation
- ROS and ROS 2
- Motion capture (MoCap)-based experimental systems

Programming & Computing:

- C, C++
- Python
- MATLAB / Simulink